

BIVARIATE RANDOM VARIABLES

We extend the concept of random variable to 2 dimensions

bivariate random variable $(X, Y): \Omega \rightarrow \mathbb{R}^2$

- The CDF is now a function $F_{(X,Y)}: \mathbb{R}^2 \rightarrow [0,1]$

$$F_{(X,Y)}(x,y) = \mathbb{P}((X,Y) \in (-\infty, x) \times (-\infty, y)) = \mathbb{P}(X \leq x, Y \leq y)$$

- discrete r.v.'s:

$$\text{joint probability function } P_{(X,Y)}(x,y) = \mathbb{P}(X=x, Y=y)$$

$$\text{marginal probability function } P_X(x) = \mathbb{P}(X=x) = \sum_{y \in S_Y} \mathbb{P}(X=x, Y=y)$$

- continuous r.v.'s:

$$\text{joint density function } f_{(X,Y)}(x,y)$$

$$\text{marginal density function } f_X(x) = \int_{-\infty}^{+\infty} f_{(X,Y)}(x,y) dy$$

- INDEPENDENCE: Two r.v.'s X and Y are independent ($X \perp Y$)

$$\Leftrightarrow f_{(X,Y)}(x,y) = f_X(x) \cdot f_Y(y) \quad (\text{continuous case})$$

$$\Leftrightarrow P_{(X,Y)}(x,y) = P_X(x) \cdot P_Y(y) \quad \text{i.e. } \mathbb{P}(X=x, Y=y) = \mathbb{P}(X=x) \cdot \mathbb{P}(Y=y) \quad (\text{discrete case})$$

- COVARIANCE between X and Y : $\text{cov}(X,Y) = \sigma_{XY} = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$

it expresses how the two variables change together

- CORRELATION: $\text{corr}(X,Y) = \rho_{XY} = \frac{\text{cov}(X,Y)}{\sqrt{\text{var}(X) \text{var}(Y)}} \in [-1,1]$

We can extend these concepts to a generic dimension $d \geq 1$.

MULTIVARIATE RANDOM VARIABLES (RANDOM VECTORS)

A multivariate r.v. is a column vector $\underline{X} = [X_1 \ X_2 \ \dots \ X_d]^T$ whose components are r.v.'s

$[X_1 \ \dots \ X_d]^T: \Omega \rightarrow \mathbb{R}^d$

- CDF $F_{\underline{X}}: \mathbb{R}^d \rightarrow [0,1]$

$$F_{\underline{X}}(\underline{x}) = \mathbb{P}(X_1 \leq x_1, X_2 \leq x_2, \dots, X_d \leq x_d)$$

- EXPECTED VALUE: $\mathbb{E}[\underline{X}] = \begin{bmatrix} \mathbb{E}[X_1] \\ \vdots \\ \mathbb{E}[X_d] \end{bmatrix}$ d -dimensional vector

- COVARIANCE MATRIX $\text{var}(\underline{X}) = \mathbb{E}[(\underline{X} - \mathbb{E}[\underline{X}])(\underline{X} - \mathbb{E}[\underline{X}])^T] =$
 $= \mathbb{E}[\underline{X}\underline{X}^T - \underline{X}\mathbb{E}[\underline{X}]^T - \mathbb{E}[\underline{X}]\underline{X}^T + \mathbb{E}[\underline{X}]\mathbb{E}[\underline{X}]^T] =$
 $= \mathbb{E}[\underline{X}\underline{X}^T] - \mathbb{E}[\underline{X}]\mathbb{E}[\underline{X}]^T - \mathbb{E}[\underline{X}]\mathbb{E}[\underline{X}]^T + \mathbb{E}[\underline{X}]\mathbb{E}[\underline{X}]^T =$
 $= \mathbb{E}[\underline{X}\underline{X}^T] - \mathbb{E}[\underline{X}]\mathbb{E}[\underline{X}]^T \Rightarrow d \times d \text{ matrix}$

what are the elements of this matrix?

$$\mathbb{E} \left[\begin{bmatrix} X_1 - \mathbb{E}[X_1] \\ X_2 - \mathbb{E}[X_2] \\ \vdots \\ X_d - \mathbb{E}[X_d] \end{bmatrix} \begin{bmatrix} X_1 - \mathbb{E}[X_1] & X_2 - \mathbb{E}[X_2] & \dots & X_d - \mathbb{E}[X_d] \end{bmatrix} \right]$$

$$= \mathbb{E} \left[\begin{bmatrix} (X_1 - \mathbb{E}[X_1])^2 & (X_1 - \mathbb{E}[X_1])(X_2 - \mathbb{E}[X_2]) & \dots & (X_1 - \mathbb{E}[X_1])(X_d - \mathbb{E}[X_d]) \\ (X_2 - \mathbb{E}[X_2])(X_1 - \mathbb{E}[X_1]) & (X_2 - \mathbb{E}[X_2])^2 & \dots & (X_2 - \mathbb{E}[X_2])(X_d - \mathbb{E}[X_d]) \\ \vdots & \vdots & \ddots & \vdots \\ (X_d - \mathbb{E}[X_d])(X_1 - \mathbb{E}[X_1]) & (X_d - \mathbb{E}[X_d])(X_2 - \mathbb{E}[X_2]) & \dots & (X_d - \mathbb{E}[X_d])^2 \end{bmatrix} \right] =$$

$$= \begin{bmatrix} \text{var}(X_1) & \text{cov}(X_1, X_2) & \dots & \text{cov}(X_1, X_d) \\ \text{cov}(X_2, X_1) & \text{var}(X_2) & \dots & \text{cov}(X_2, X_d) \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}(X_d, X_1) & \text{cov}(X_d, X_2) & \dots & \text{var}(X_d) \end{bmatrix} = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \dots & \sigma_{1d} \\ \sigma_{21} & \sigma_2^2 & \dots & \sigma_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{d1} & \sigma_{d2} & \dots & \sigma_d^2 \end{bmatrix} \quad \begin{array}{l} \text{symmetric} \\ \text{positive semi-definite} \end{array}$$

MULTIVARIATE NORMAL DISTRIBUTION

generalization of the normal distribution to d dimensions

$$\underline{X} = [X_1 \ \dots \ X_d]^T \sim N_d(\underline{\mu}, \Sigma)$$

- support $S_X = \mathbb{R}^d$

- parameters:

- expected value $\underline{\mu} = \mathbb{E}[\underline{X}] = [\mathbb{E}[X_1] \ \dots \ \mathbb{E}[X_d]]^T$ d -dim vector

- covariance matrix $\Sigma = \text{var}(\underline{X})$ $d \times d$ matrix

- density function

$$f_{\underline{X}}(x_1, \dots, x_d) = (2\pi)^{-d/2} \det(\Sigma)^{-1/2} \exp \left\{ -\frac{1}{2} (\underline{x} - \underline{\mu})^T \Sigma^{-1} (\underline{x} - \underline{\mu}) \right\}$$

- marginal distributions: example $[X_1, X_2, X_3]^T \sim N_3(\underline{\mu}, \Sigma)$ $\left(\underline{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix}, \Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_2^2 & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_3^2 \end{bmatrix} \right)$

the marginal distributions are simply obtained by looking only at the components we are considering

e.g.

$$X_1 \sim N_1(\mu_1, \sigma_1^2)$$

$$[X_1, X_2]^T \sim N_2 \left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{bmatrix} \right)$$

MULTIVARIATE STANDARD NORMAL

$$\underline{\mu} = \underline{0} \quad \text{and} \quad \Sigma = I_d \quad \underline{Z} \sim N_d(\underline{0}, I_d)$$

in this case, the Z_i ($i=1, \dots, d$) are independent normal r.v.'s $Z_i \sim N(0,1)$

- general normal $\underline{X} \sim N_d(\underline{\mu}, \Sigma)$ from the standard normal $\underline{Z} \sim N_K(\underline{0}, I_K)$

$\underline{\mu} \in \mathbb{R}^d$ d -dim vector, A $d \times k$ matrix such that $\Sigma = AA^T$

$$\underline{X} = A\underline{Z} + \underline{\mu} \Rightarrow \underline{X} \sim N_d(\underline{\mu}, \Sigma)$$

1) linear transformation of a normal r.v. is normal

$$2) \mathbb{E}[A\underline{Z} + \underline{\mu}] = A\mathbb{E}[\underline{Z}] + \underline{\mu} = \underline{\mu}$$

$$3) \text{var}(A\underline{Z} + \underline{\mu}) = \text{var}(A\underline{Z}) = \mathbb{E}[(A\underline{Z} - A\mathbb{E}[\underline{Z}])(A\underline{Z} - A\mathbb{E}[\underline{Z}])^T] =$$

$$= \mathbb{E}[A\underline{Z}\underline{Z}^T A^T - A\underline{Z}\mathbb{E}[\underline{Z}]^T A^T - A\mathbb{E}[\underline{Z}]\underline{Z}^T A^T + A\mathbb{E}[\underline{Z}]\mathbb{E}[\underline{Z}]^T A^T] =$$

$$= A\mathbb{E}[\underline{Z}\underline{Z}^T]A^T - A\mathbb{E}[\underline{Z}]\mathbb{E}[\underline{Z}]^T A^T - A\mathbb{E}[\underline{Z}]\mathbb{E}[\underline{Z}]^T A^T + A\mathbb{E}[\underline{Z}]\mathbb{E}[\underline{Z}]^T A^T =$$

$$= A(\mathbb{E}[\underline{Z}\underline{Z}^T] - \mathbb{E}[\underline{Z}]\mathbb{E}[\underline{Z}]^T)A^T$$

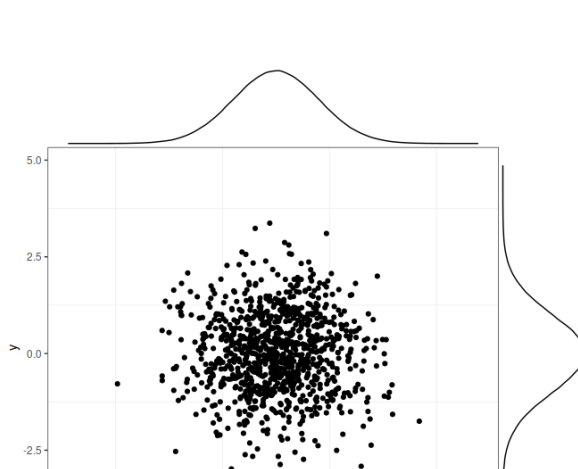
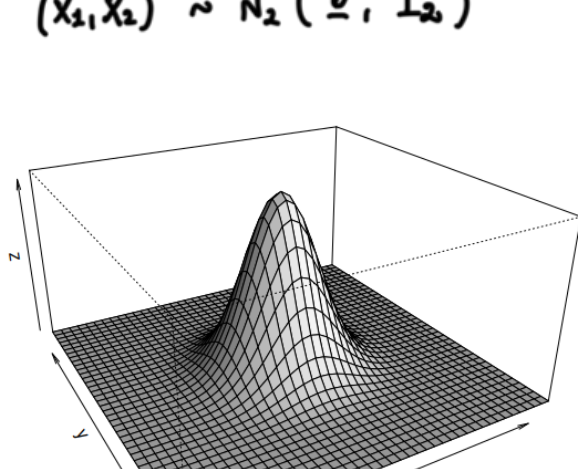
$$= A \underset{I_K}{\text{var}(\underline{Z})} A^T = AA^T = \Sigma$$

BIVARIATE NORMAL ($d=2$)

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim N_2 \left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{bmatrix} \right)$$

EXAMPLES

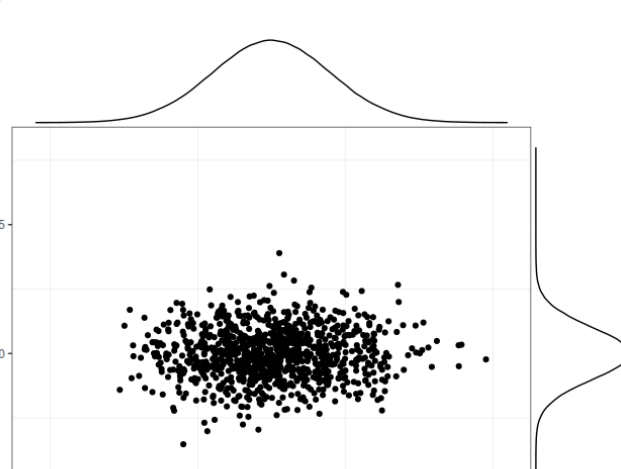
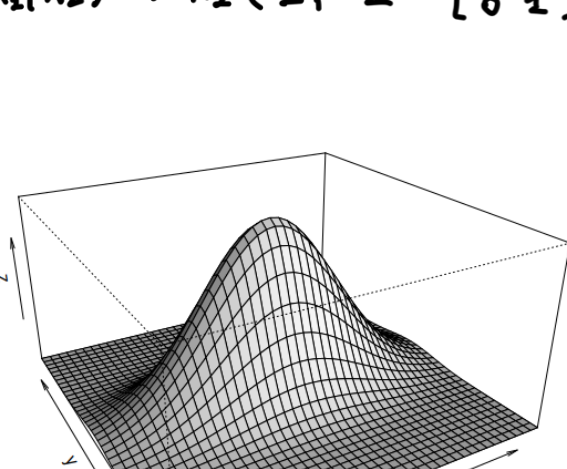
$$(X_1, X_2)^T \sim N_2(\underline{0}, I_2)$$



$$X_1 \sim N(0,1) \quad X_2 \sim N(0,1)$$

$$[X_1, X_2]^T \sim N \quad \text{and} \quad \text{cov}(X_1, X_2) = 0 \Rightarrow X_1 \perp X_2$$

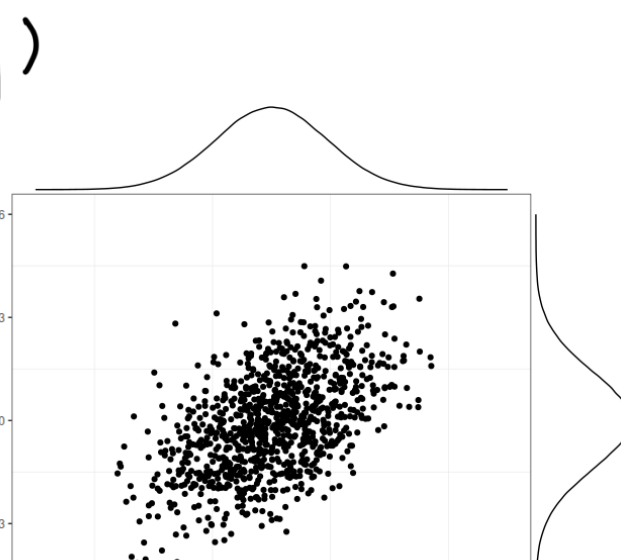
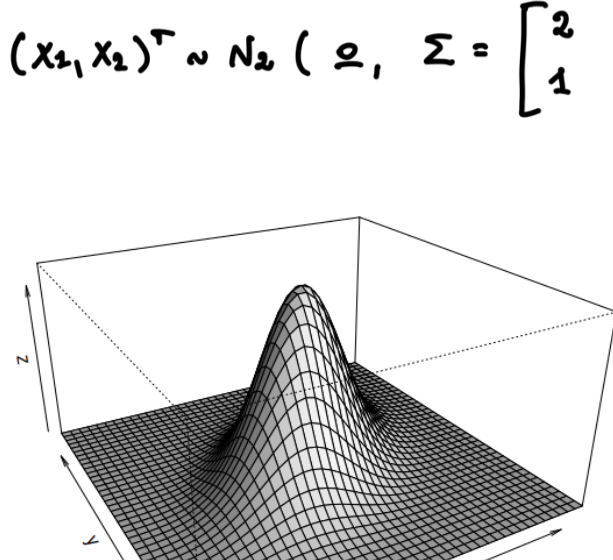
$$(X_1, X_2)^T \sim N_2(\underline{0}, \Sigma = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix})$$



$$X_1 \sim N(0,4) \quad X_2 \sim N(0,1)$$

$$X_1 \perp X_2$$

$$(X_1, X_2)^T \sim N_2(\underline{0}, \Sigma = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix})$$

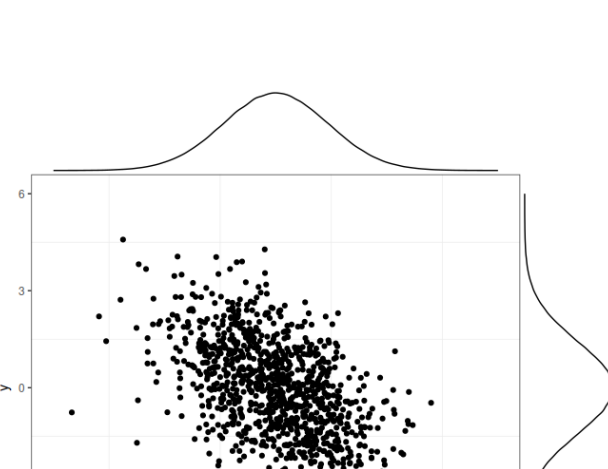
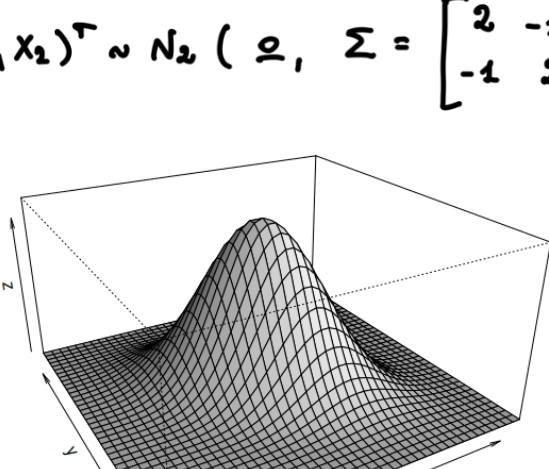


$$X_1 \sim N(0,2) \quad X_2 \sim N(0,2) \quad \text{same variance}$$

they are not independent!

positive correlation: at large values of X_1 we expect large values of X_2

$$(X_1, X_2)^T \sim N_2(\underline{0}, \Sigma = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix})$$



$$X_1 \sim N(0,2) \quad X_2 \sim N(0,2) \quad \text{same variance}$$

they are not independent!

negative correlation: at large values of X_1 we expect small values of X_2